

Phase 1 paper session

Everything physical the project needs, in one list. Nothing else is being asked for before this, and nothing has been held back to ask for later.

Split into three sittings so the range day stays short. Only part B needs the range, and part A can be done any evening.

- **Part A, at home, before you go.** Print everything, scan what never gets shot, scan the sheets you are about to shoot. About 45 minutes.
- **Part B, at the range.** Shoot two sheets, photograph them mounted before you take them down. About 15 minutes on top of a normal session.
- **Part C, at home, after.** Scan the shot sheets, drop the files in folders. About 20 minutes.

About 50 rounds. Two sheets, one shot per bull. A third sheet is optional and only if you want a second calibre in the set.

Part A: at home, before you go

A1. Print the pack

Ask Claude Code to generate it:

Produce the Phase 1 print pack as PDFs in `scans/phase1/print-pack/` : three copies of the corrected `GL-CF25-LTR` from the geometry commit, and the five marker module sweep sheets from `scans/phase1/module-sweep/` . Print one page each, Letter, and tell me the identifier printed on each sheet so I can check the right geometry went out.

Eight pages. Three target sheets and five sweep sheets.

Print settings, and the one that matters. Brother MFC-J430W, plain paper, Normal quality, portrait, and **scale set to 100 percent with "fit to page" and "shrink to fit" both off**. Everything else in this session is worthless if the scale is wrong, and a printer driver will silently shrink a page to its own margins given the chance. These are the settings from the Phase 0 protocol, unchanged, so the new sheets stay comparable with the old ones.

Check one sheet before printing the rest. Measure between two bull centres across the widest span with the calipers. It should be five grid pitches. If it is short by a percent or two, scaling is on somewhere.

A2. Label them, briefly

One code per sheet, small, in pen, in the bottom margin. Nothing else. The filename carries everything else.

Code	Sheet	What happens to it
S1	GL-CF25-LTR	Shot normally
S2	GL-CF25-LTR	Shot deliberately awkwardly
S3	GL-CF25-LTR	Optional, second calibre
C1	GL-CF25-LTR	Never shot. Flat control
M03 M04 M05 M06 M08	Sweep sheets	Never shot. Scanned only

Last time labelling and scanning cost you an hour. This is one code per sheet and the codes are two characters.

A3. Scan the five sweep sheets

600 DPI, colour, no auto-crop, no descreen, no sharpening. Save as `scans/phase1/module-sweep/scans/M03-600.jpg` and so on.

These never go to the range. They exist to find the smallest marker your printer can lay down cleanly, which is the one measurement nobody can make without your actual printer.

A4. Scan S1, S2 and S3 before you shoot them

Same settings. Save as `scans/phase1/paper/S1-before-600.jpg` and so on.

This is the step most worth not skipping. Your printer puts ink about five hundredths of a millimetre from where it is asked, and that error travels with the paper rather than with the printer. Comparing a shot sheet against a scan of that same sheet removes it completely. Comparing against the ideal design does not. Ten minutes of scanner time buys a cleaner answer than any amount of software.

A5. The flat control, C1

Lay C1 flat on a table indoors. Photograph it square on, whole sheet, all four edges in frame with a bit of margin, in decent even light. Two or three frames is plenty.

Then scan it, `C1-600.jpg`.

This is the only sheet in the set printed from the corrected geometry and never shot, and it answers a question that is currently open: whether the sighter row fix actually makes the flat photograph gate pass. Right now that is recorded as not measured.

Take one of these frames carefully. Square on, well lit, whole sheet, nothing clipped. That one gets used as the "yes, like this" example on the upload page, replacing the drawing I made because no such photo exists anywhere in the project.

Part B: at the range

B1. Shoot S1 normally

Your real load, the way you would actually shoot it. 25 rounds, one per bull. Nothing special.

B2. Shoot S2 deliberately awkwardly

This is the more valuable sheet and the easier one to shoot. A second sheet of well-centred groups teaches the software nothing the first one did not. Spread the holes on purpose:

- Several on bare paper between bulls, well away from any printing
- Several cutting through a printed ring stroke
- A couple on or touching a printed numeral
- **At least one pair overlapping inside a single bull**, close enough that the two tears merge
- **At least one shot landing in a neighbouring bull's cell**

25 rounds. Do not try to make it neat.

B3. Optional: S3, a second calibre

Only if you want hole diameter variety in the set. Hole size tracks bullet diameter, so .308 on one sheet and 6.5 mm on another spans the range. Skip it without guilt.

B4. Photograph each shot sheet while it is still mounted

Before you take anything down. This cannot be recreated afterwards, and it is the case the project has never had: a whole GroupLab sheet, with its markers, still fixed where it was shot.

Four frames per sheet, thirty seconds each:

1. Square on, from wherever you would normally stand to take the photo
2. Deliberately off to one side, a steep angle
3. Further back, so the sheet is smaller in the frame
4. Whatever the light is doing worst. Your shadow across it, or sun glare, or deep shade

Take them **however you normally photograph a target**. Do not try to take good photographs. Then take one extra frame of each sheet with the main camera specifically, so there is something directly comparable with the Phase 0 set.

Whole sheet, all four edges in frame, every time. A close-up of the group is the one thing that cannot be used.

B5. Two marks in pen, on S2 only

After shooting, before it comes down. Thirty seconds:

- Draw an **X** over one hole, the way you would to exclude it
- Draw an **arrow** from a bull to the shot that landed in a neighbour's cell

Real targets in the sample set carry both, drawn by whoever shot them, and the software has to cope. This is the first time either will exist on a sheet whose geometry is known exactly.

Part C: at home, after

C1. Scan the shot sheets

600 DPI, same settings as before. S1-after-600.jpg, S2-after-600.jpg, S3-after-600.jpg.

If a sheet is torn or curled, scan it anyway and do not flatten it first. A damaged sheet is a real case.

C2. Put everything where it goes

```
scans/phase1/paper/      S1/S2/S3 before and after, C1
scans/phase1/mounted/    the range photographs
scans/phase1/module-sweep/scans/  M03 to M08
```

Then commit and push.

C3. Tell me three things

1. Anything that went wrong or felt stupid. Especially if a step was more work than it looked.
2. How the scale check in A1 came out.
3. Whether the printer produced anything visibly rough on the 0.3 mm sweep sheet, which is the one most likely to be past what it can do.

What this produces, and why each piece is here

Item	Answers
S1, before and after	Hole detection on known geometry, against a scan of the same physical sheet
S2, before and after	The hard cases: overlapping holes, holes on ink, cross-cell shots, pen marks
Mounted photographs	The mounted gate, which currently fails on nine frames of one sheet hanging from a single pin. This is the first realistic sample
C1, flat and scanned	Whether the corrected sighter geometry passes the flat gate. Currently unmeasured
One careful C1 frame	Replaces the drawn example on the upload page
Sweep sheets M03 to M08	The smallest marker your printer can lay down, which no literature can supply

Nothing here is being kept back for a second session. If something in this list turns out to be wrong or missing, that is my error rather than a planned follow-up.